

<https://www.youtube.com/watch?v=xS3nE81Cm7E>

Hi,

In this video I will be introducing
Kahneman
and try to explain what question he's
actually answering what topic is he trying
to deal with here.

So usually the focus of philosophy of
science
is on research designs, on what a theory
actually is, what methods we should use,
and what social world we are actually
interested in.

What we're doing here with the inclusion
of this text is to say well we should

also consider
the human beings that both carry out the
research

so human beings and researchers but it's
also human beings that we
do economic uh research on so

who are these human beings and what
characterizes them let's see
essence of what we are trying to go
through here is we'll be trying to
understand what human rationality
actually
is

Now, consider this. Usually classical economics has assumed: **“everyone is reacting rationally to the market situation”** (quote by Spence, Nobel Prize winning economist)

In contrast, Kahneman questions if this assumption about a rational homo economicus really makes sense. He wants to explore and explain how does the human mind ACTUALLY work and how do humans actually behave? He highlights how social scientists in the 70s broadly accepted two ideas about human nature: first, people are generally rational, and their thinking is normally sound. Second, emotions such as fear, affection and hatred explain most of the occasions on which people depart from rationality. Kahneman

wants to challenge both these assumptions. He wants to document the systematic errors in thinking of normal people, and traces these errors to the machinery of cognition, rather than corruption of thought by emotions.

So, really important to highlight, a core issue, that Kahneman wants to shift our assumptions about why we sometimes make mistakes. But, also, we should at the end get back to why these biases might be a good thing at a collective, long term level and perspective.

First, what really characterizes our thinking: biases and the distinction between system 1 and system 2. This is really key to layout.

You might have heard the term bias:
that we human beings are
biased- what does that mean and where
do these biases come from
and then also why are these
important, and this links to
the next lesson about Kuhn as well so
where does irrationality of thinking
come from we'll be getting a sense of
system one and system two in the next
video
and then why do we have these cognitive
biases that you've heard about
and what implications does this have.

Where Does Irrationality Come From?

I'll try to explain what this means by
providing a couple of examples: So
this is a classical example that
behavioral economics is basically all
about
so here you see a question that says
check the box if you want to participate
in the pension
scheme or whatever sort of organ
donation or whatever the question could

be

And then here you have check the box if you don't want to participate in the pension scheme.

The content of the question is the same:

“Do you want to participate?” yes or no

So in that sense whatever you sort of do

whether you click the box or not

it's answering the same

question, it's trying to get to the same

essence but it turns out that

we get quite different answers

depending on how we actually

frame this question and this doesn't

quite match with this idea of homo

economic as this

classical economic idea about fairly

rational human beings that process

information and then sort of

express their preference. According

to that

so this is just one example and

I'll provide a few more which are

meant to illustrate that we, human

beings, are not as irrational as we

might like to think ourselves to be we

should be answering: all human beings

if they were rational should answer this

question in the same way

the same number of people that sort of

the reverse number of people that take

the box here should

then not take the box in the next one so

but it is the case and we see lots of

examples and richard taylor has basically

got a nobel prize in economics for

demonstrating this

so the the question we are asking here

that we are focusing on here with

kahneman is

where does this sort of seeming

irrationality

where does this irrationality seem to

stem from and there are basically two

kinds of answers we could provide

it's either because emotions are taking

over, it's simply because we are emotional when we should be rational another angle could be well our brains are just not working, they're just inefficient but very crudely put two kinds of responses we could provide to the question where does this irrationality come from



Princeton vs Dartmouth game: Emotional Influence?

and an example such as this one might sort of fit this idea of emotional being so this is a very classical example many others like this you probably know about there was a football game going on and it was a very intense game and it ended out in a fist fight. The fist fight erupted ,there were further injuries on both princeton and dartmouth side and then what was done was that there were two psychologists that took videos of this game and showed them to students on each campus and each student was supposed to note how this all happened and what sort of led to the fist fight and try to find out who was responsible for it and and i'm sure you're not surprised about that princeton students generally saw twice as many dartmouth violations as the other way around so they were watching the same video but they ended up with different conclusions about who was responsible for starting these fist fights who would actually sort of violate , made the most violationsso even though there is sort of an objective reality that we could fill we have different opinions on it i



think here we might tend to think well
it's just because they have emotional
attachments to one of the football teams
so
they didn't really pay as much attention
as they should do

Wheel of Fortune: Anchoring bias

but when we then consider this example
which is an example **Kahneman has used**
often



we might become a little bit more
confused about why
are we being this irrational so imagine
an experiment two different groups
that both get the question what
percentage of **UN** members are african
countries
but before each of these groups would
answer the question
a wheel of fortune was spun right before
them and either it stopped at 10
or it stopped at 65.
and in the group where it stopped at 10
the mean answer
to the question of what percent of even
members are african countries
was 25% well in the one where the
number stopped at 65 the mean answer
was 45% so this is an example where
there
clearly does not seem to be any
emotional attachment involved we
we don't sort of care about the number
on the wheel of fortune or
the number of African countries in in
sort of a very
deep way but it seems to be the case
that when we see a very low number
we end up answering a lower number **than**
if we see a higher number
so maybe there is something about each
group of people they sort of know that
the answer is somewhere between 25 and
45 but depending on how the question and
answers
is asked we get different answers

and there are a few more examples we can show of this this is also an example

The Example of Linda (reflection exercise)

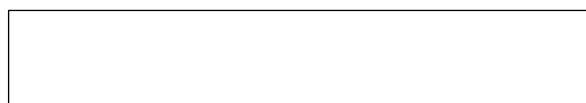
that Kahneman is using
so think of this person Linda here you can stop the video and just read it also if you want to
so Linda is this 31 year old single outspoken very bright woman major in philosophy as a student she was deeply concerned with issues of discrimination and social justice and even participated in anti-nuclear demonstrations
now which alternative is the most probable
which of these three descriptions is likely to fit Linda you can pause the video reflect on it
but the key thing really to note here is if we're thinking about probability well **Linda is a bank teller** in this scenario she's both a **bank teller** and active in the feminist movement well in any case the bank teller angle is most likely you can sort of see it in in a Venn diagram here
that any time she's a feminist bank teller she's also a bank teller so the answer you should give is Linda is a bank teller but it turns out Kahneman has carried out this exercise in many places around the world
fancy universities with statisticians with students with people in general and people tend to end and so this is a bank teller active in the feminist movement
and uses this example in the book as well so we
again this could sort of be an indication of a brain that is not 100

rational because if we're thinking of
pure probabilities
the question is really not that
difficult but that's not
what our brain tends to do
you'll see more of **kind of** of examples
throughout these two videos



Irrationality: Cognitive challenge

but it really gets to the question of
where does this irrationality of
thinking come from which he talks about
here on page and he says that social
scientists in the 70s have broadly
accepted two ideas: either
for one people are generally irrational
and their thinking is normally sound
and second emotions such as fear
affection hatred etc.
can make people depart from rationality
so the basic idea
is that we are sort of information
processing machines just like a computer
and emotions might sort of make our
brains go so
whatever sort of flawed answers we would
have given to the three four examples
we've seen here in this video
it's because of emotional issue but as
he then says
he's trying to challenge these
assumptions about
human nature and the brain and they're
trying to document systematic errors in
the thinking of normal people
and we trace these errors to the design
of the machinery of cognition that is
our thinking
rather than to the corruption of thought
by emotion
so the idea is to sort of make us
understand that we should not think
ourselves as these **shore economic as**
the information processing machine that
is rational long as emotions don't



intervene this is the
key idea that he's trying to go against
in the book
but also that he's been trying to
promote that we have these systematic
errors and it's a
cognitive issue it's a rationality
issue he's developed i mean as you can
see he wrote some of the first papers
here in the 70s and he since then
developed
his work he's a psychologist and then
got the nobel prize in economics in in
2002 and then also wrote a book called
thinking fast and slow in 2011 which
has really
made him i mean quite famous it's an
excellent book and it's the one that
we're reading a couple of chapters from
where you can also see we sort of look
at
the number of times that the word
behavioral economics so
how people actually behave so what is
people's actually economic behavior if
we see how often this term is used in
economics papers you can see that there
really something was going on around the
year 2000 and
onwards it's just used a lot more so we
are
more interested in the economics
literature about
how do people actually behave rather
than just assuming this formal economic
that it is
rational and focused on monetary
incentives



Summary and Next Lesson

so that's the question that that
kahneman is dealing with here is and
this is why we're drawing upon this in
this philosophy of science course
yes we want to know about research
designs and methods and what a theory
actually is and what social world are we

even doing research about
but we can't forget the people that are
doing the research
and the people that we actually do the
research on what
what is what are these human beings and
a typical
classical economic assumptions this is
by an article by spencer once wrote :
everyone is reacting rationally to the
market situation that was the
assumptions when they were analyzing
data on how humans behaved and kahneman
is then basically saying we should not
think of the brain as this
information processing machine we have
two different systems
two different cognitive systems two
different thinking systems
and this is what is then trying to
explain how does a human mind
actually work that's what systems one
and system two is about that's what
is being presented in the book chapters
and that's what i will be getting into
in the
next video



<https://www.youtube.com/watch?v=QOOdtPJlIbk>

hi in this video i will be presenting
kahneman's distinction between
system one and system two thinking
so system one is
a kind of thinking that operates
automatically and quickly with little or
no effort
and no sense of voluntary control it's
looking at this picture and immediately
realizing that there is some kind of
negative emotion
she's portraying being angry or she
seems to be yelling. we don't
need to look at the picture or think
about the position of the mouth and
how open it is and what do the eyes
actually look

like and how about the eyebrows
it's immediate, and it's automatic -
it's quick
and there's not really an effort going
on
same with two plus two that equals four
people with sort of a regular danish
education
don't need to stop and think about this
equation we can look at the numbers here
and say
12 13 14 again it's easy,
it's automatic and then it would even be
difficult
not to sort of read it or
be able to figure out that this two plus
two is four
I can say the words pink elephant and
it's difficult not to think of them
as a of a pink elephant it's difficult not
to sort of
engage in this quick system one thinking
in contrast
if we have to do a slightly more
complicated
piece of math it becomes more
difficult it's not that we can't do it
10 times 24 is 240, 7 times 24
is 168
and then we can add these two numbers
but as you can see i really had to
focus and concentrate on that system two
thinking
is a kind of thinking that requires an
allocation of attention
it's an effort for mental activity for
instance and it's some kind of
complex competition it doesn't have to
be math it can be any
other kind of more complicated thing and
system two operations
are then often associated with sort of
the feeling of agency and choice you
have to decide to try to figure out what
17
times 24 is it's not like that two plus



two
is four you have to actively engage in a
kind of thinking you have to concentrate
you might even sort of experience if you
are walking with someone and then
suddenly and
a difficult question is being asked the
other person might even stop walking in
order to think about the question
it requires such intense concentration
that we sort of end up
stopping doing anything else so
kahneman'
very fundamental point is here that this
is both
both of these systems that the kinds of
operations they illustrate and portray
are cognitive processes this is
something that happens in our brain
it is a kind of thinking but they are
fundamentally different
and one is automatic and quick and the
other one requires an active engagement
it
requires concentration
and can also be exhaustive you can get
tired of having to concentrate
for too long so this is again just to



The Rabbit/Turtle Metaphor

sort of provide an overview of
what we could call the turtle versus a
rapid kind of thinking and there are a
bunch of examples here on page 21 to
22
and what this system one kind of
thinking can be
being able to hear where sound is coming
from
detecting whether someone is hostile
understanding simple sentences
looking out into the horizon and seeing
that one object is more distant than the
other
it's quick, it's effortless, it's

automatic it's almost difficult not to
do it
in contrast if there is a noisy room and
we have to sort of find out who's
actually talking right now
we really have to concentrate and we
have to pay attention to it
if you hear a surprising sound that
you're not really sure you're familiar
with then you really have to search your
memory
again you have to engage in an effortful
process of a mental activity
if you're reading a difficult text or
sort of considering whether someone's
argument is really sound and valid well
again you
have to engage in this system two kind
of thinking and then again there are
further examples on page 21 to page 22.
so system one is
can also then be characterized as
gullible because we can very quickly
sort of
this is where we jump to conclusion
while system two is then in charge of
being doubtful and not necessarily
trusting what we hear
but system two is lazy and then often
busy and because it requires this
attention and effortful activity then we
don't always engage system two when we
should and again that's really a key
point
i'll be trying to drive home a couple of
times in this lesson
and that system one is what leads us to
have
biases for instance confirmation biases
and jumping to conclusions that maybe we
shouldn't jump to
it's our individual responsibility
to engage system 2 to stop and to think
about what we are being faced with
but it requires an effort and attention
of course we can then sort of consider
now i used examples two plus two and

seventeen times twenty four i mean where
do we sort of
draw the line when does it move from
system one to system two there was no
clear cut off and it certainly depends
on how much experience one has with math
and i'm sure there are people that can
do 17 times 24 immediately because
they've just done so much math in their
life so just think of these as
sort of principles and illustrative
examples
of two different kinds of thinking
processes and they're not two completely
different separate systems in our mind
it's probably a continuum between system
one and system two but they still
represent
different kind of cognition and
different kinds of thinking and and
we'll get many more examples and
implications of this in the in the
following

The Stroop Test - Simple but Effortful

another example that can illustrate how
engaging in system two thinking can be
effortful is by doing what we can call
this group test you can even google this
and then you can
find sort of various tests we can try to
engage in it but it's basically
looking at words and colors like these
and then having to say the color
not read the word red but say the
color that is underneath it this is then
red
and white and green and brown so that's
quite easy
but when we then have to read this one
as red and this one is white and this
one is red and this one
has green and white, brown
it actually requires an effort and try
to find one of these tests online
and do this for a minute or two in a row
it becomes really cognitive demanding

and very very difficult to continue to
engage in even though the task in itself
is extremely easy
it's just about looking at the color of
a word
and saying that color but but that's
again just to illustrate that
that even fairly simple tasks can become
sort of
cognitively demanding so the difference
between system one and system two is not
per se easy versus difficult
but it is effort for versus effortless
it is quick
versus slow that's why the book is
called thinking fast
and thinking slow but the point here is
also that
system one is sort of eager to just jump
to a conclusion of saying
reading white and green and red etc.
even though that's not what we're
supposed to read so system one is always

Jumping to Pre-existing Conclusions

ready just to jump to
whatever conclusion that it sort of
immediately springs to our mind and
a little bit of a complicated setup is
this is trying to
to illustrate this point
where this is an experimental setup
where someone has
put out a bunch of these moving dots and
it's
and again it's an experiment imagine two
different groups
and they see two this sort of c these
all these moving dots in two rounds
in the first round they have to figure
out whether the dots generally move to
the left or to the right
and then they have to make a decision on
that and
in the second round these dots are again
all moving to the left
to the right and as highlighted in the

bullet points here to the left
what happens is that if you in the first
round saw the dots move to the left
you're much quicker to be able to
identify that in the second round also
move to the left
so the task in itself is of course
meaningless in many ways but it's
it's an effective task to figure out
that
our brain is very eager and quick to
jump to a conclusion that it sort of
already jumped to before
and the point that is that we might sort
of be able to extrapolate that to
real-life cognitive processes as well
so people's interpretation of new
evidence is often biased by the previous
choice as highlighted here in the first
bullet point
if we've seen something before we are
much more eager to
sort of see that same thing again in the
second round
system one makes us ready to process
certain information

Overview of Systems 1 & 2

so again just an overview that
highlights the different characteristics
of
system one and system two system one
it's unconscious it's effortless it's
automatic there's no self-awareness
what you see is all sort of all there is
and the role of system one isn't to
deliver updates assess the situation and
just sort of continuously
informing us as individual what's going
on right now
system two is then the slow thinking it
requires a conscious deliberate effort
and requires us to decide to engage in
a thinking effort it's the part of us
that is logical and skeptical
and might say let's try to search
for more information i'm not really sure

about this
it's also the one that then is supposed
to make decisions
again these are sort of simplified
distinctions
and they're not sort of two different
places in the brain if not the left side
or the right or the front or the back
side
but it's sort of characterizations of that
our brain is not just
one thing we actually engage in two
different kinds of
thinking there are two different kinds
of
thinking processes going on in our mind

"Whodunnit" clip

and system one can end up filtering what
we get
let's watch this short video that makes
this illustration sort of a typical
who done it who's responsible video let's
just watch this for
about a minute a minute and a half
clearly somebody in this room murdered
lord smythe
who at precisely 3:34 this afternoon
was brutally bludgeoned to death with a
blunt instrument
i want each of you to tell me your
whereabouts
at precisely the
i was polishing the brass in the master
bedroom
i was buffering his lordship skulls
below stairs
sir well i was planting my petunias in
the potting shed
possible arrest lady smith
but how did you know madam as any
horticulturist will tell you one does
not plan petunias until may is out take
her away
it's just a matter of observation the
real question



is how observant were you
somebody in this room murdered lord
smythe who
at precisely 3:34 this afternoon was
brutally bludgeoned to death with a
blunt instrument
i want each of you to tell me your
whereabouts
at precisely the time that they started
took
place i was polishing the brass in the
master bedroom
i was buttering his wanted scones the
host stairs
son but i was planting my petunias in
the potting shed
possible arrest lady smith



Top-down Filtering, Information and Attention

so a sort of a simple
fun example that illustrates that we
might
think that we see the world and that we
process
the information that is being put before
us but it's often just system one that is
engaging in sort of
effortless thinking and then focusing on
sort of
whatever is at interest here and the
main question was who actually did it
who did the murder
and it's system one filter so system two
processes
sort of the logical part of the brain
can only sort of
deal with information that is actually
being presented to system two
and the point is here then also that
that we might think that we have all
information in front of us
but since system one sort of is the
effortless filtering that
um filters whatever information is
coming from the outside and what system
two can even work with when it's trying
to figure out who actually did it

well that leads to a problematic situation and we have to realize that we haven't necessarily seen everything that is that is going on and so we have this top-down filtering that the formal video illustrated and these examples also illustrate as you can clearly see on the left it says abc and here it says 12 13 14 but notice that the b and the 13 are completely identical but our brain is very eager and quick to just neglect the ambiguity here and say abc and 12 13 14 because we have this top-down filtering going on system one is just processing the information it's not really thinking about what's going on here it's just saying oh look here's an a here's a c that's probably a b in the middle um and then the next example is also quite fascinating when we look at this picture it's a black and white picture is only the lines that are colored so everything sort of within these boxes here is not colored but our brain is just eager to sort of add more information to whatever we're seeing so we're seeing the picture and color on unless you are blind of course but but so and so our brain is not really capable of sort of making these distinctions that it's only the lines that are colored and the rest is black and white so our brain is quite literally a prediction machine that's also what happened in the video before in and whodunnit video our brain is filtering and helping us sort of

expect what to see which then also means
that we don't sort of see
everything that is put before our eyes
quite literally
this is also illustrated in this in this
visualization
so if you sort of reach out your arm
and look at the size of your thumb
that's really the
extent of the very high quality
information you're getting in visually
that's also what these pictures below
here is an illustrating that
imagine you're standing here in
santorini in Greece and you're looking
out over this village
you're actually only getting really high
quality information from
small extracts of it it's not that you
don't see
the rest but it's sort of not high
quality information you're getting in
and the reason why you sort of can see
this whole picture is in part because
you
your eyes can move around but also that
you have sort of a
background image that your brain can
pull up if you've seen this image once
then you sort of can pull that up from
your memory banks i usually when i
when i teach this in a classroom i
usually use the example that when i
walk down the
stairs in the auditorium well then i've
walked down the same stairs many times
and i
i've seen a classroom many times i
hardly have to look out into it because
my brain is already ready to
to pull out the um the image sort of
to use as a
as a frame when i engage with the
classroom
so if something was completely unusual
in the classroom i might not even notice
it even though my eyes is sort of
looking out into the room

because there is this top-down filtering
going on
this is also where you get what you can
see if you do eye tracking and the heat
map isn't illustrating what people are
actually looking at and they're not
looking at all the things on the
on the picture here some things that are
sort of really paying attention to
others they're not
so again it doesn't mean that i don't so
my it's only
the side the width of my tongue my thumb
that is providing me any kind of
information but that's where i get my
my high quality information and the rest
is really driven by
what i'm paying attention to so if i
jump back to the video this is then also
why we didn't notice all these 21
changes even though they were happening
within sort of the
in our vision and we could see the
entire screen but our brain was simply
not able to pick up all the different
elements that was going on
and we didn't notice it because system
one was just processing the information
and
the system wasn't realizing that you
should look at the picture in the
background or
all the different sort of elements that
were being changed the flowers etc
the person lying down here and the thing
that was lying there
we have a top-down filtering going on
our brain is quite literally prediction
machines and this
is also why when we look at chess
players experts versus novices
they see different things so good
chess players they
sort of see more chess pieces at once
they see larger chunks
familiar meaningful constellations of
all these different chess pieces
um which enables them to sort of more

rapidly recognize patterns
so when a novice or an expert look at a
chess board and have to tell what's
going on
they don't sort of form the same kind of
perceptions or the same kind of chunks
of what's going on they're creating
different kinds of
pictures because their systems have been
trained differently
so another example of this
is that imagine , not just imagine this
was a study
um where 100 radiologists were seeing
this picture in order to find out
whether there was
actually this was from a person who had
cancer and they're apparently especially
paying attention to all these might
different things what happened was that
83 of these
radiologists apparently did not notice
the gorilla in the picture
which i'm sure many of you noticed
because you don't really know what
you're looking for
but the point here is that what we're
thinking about what we're focused on
quite aggressively filters the world
around us and shapes what we see
and that's the effortless um unconscious
sort of system one filtering of the
world that is going on
and this is what we have to pay
attention to that we are sort of exposed
to these kinds of visual and cognitive
illusions in the world around us and
and if we're not aware of this then
system two
is working with sort of a biased version
of the actual world because they're not
getting all the
information in that we should

Summary of Lesson

so this was an attempt to provide a a

definition of what system one and system two is here you sort of have an overview of a bunch of different relevant concepts here i won't go through the details already look at it yourself either on this picture or go to the link but the the main aim was to sort of provide and a sort of initial insight into which is thinking fast and thinking slow so i've sort of covered elements from this top top left corner here um and then this top top right corner that two different kind of thinking processes they're fast and they're slow and it means the system two is difficult to engage and it's just the one that filters whatever we see in the world and and then also makes us miss all the different changes in the whodoneit video it makes us miss the gorilla it makes us think that a picture is colored because our brain is this prediction machine and it's in very sort of effective and efficient for us and it works for us then we don't always have to engage in system two imagine that we have to do this all the time every time someone says something and we have to figure out whether they're being nice or happy or hostile i imagine that we had to spend like three four seconds of really intense thinking about what is going on right here it would make life really cumbersome and problematic so it's good that we have this efficient system i'll talk more about that in the live section of this lesson but it is a point about our thinking and

this is what Kahneman isn't also saying
that we might be irrational and sort of
pick the wrong
answer in the pension scheme example
from video one here today
because it's system one that is being
engaged in in a thinking process because
we haven't stopped and
let system two sort of take over

So, be sure to make this point clear the end: Humans are rational, and when we make errors it is not just due to emotions – but because our biases have helped us create an overall effective systems, that leads to errors in particular situations. So, this ending here in the following paragraphs is really important to include.

Certainly. Biases are **practically rational** because they allow individuals to navigate complex environments where perfect information and endless time are not available. In business, decision-makers often face time pressure, incomplete data, and uncertainty. Under these constraints, trying to achieve perfect rationality would be inefficient and could lead to decision paralysis.

Instead, biases like focusing on high-status individuals or using confirmation bias allow individuals to simplify decision-making processes. These mental shortcuts help them filter vast amounts of information and focus on what's most likely to lead to success. For example, following the lead of a high-status figure or sticking with strategies that have worked before reduces the cognitive load of constantly reassessing every option.

From this viewpoint, **biases are rational because they increase effectiveness** in real-world conditions—helping individuals make timely, functional decisions that work well enough most of the time, even if they're not perfectly optimal.

Yes, this concept **does go against the traditional "homo economicus" model**, which assumes individuals are perfectly rational, fully informed, and always make decisions that maximize utility. The **homo economicus** model relies on the idea of complete information and optimal decision-making without biases. In contrast, the idea that biases are **practically rational** acknowledges that real-world decision-making involves uncertainty, incomplete information, and time constraints. Individuals use biases as efficient mental shortcuts to make good-enough decisions, rather than spending endless time seeking optimal solutions. This reflects a more realistic understanding of human behavior, as seen in **behavioral economics**, which Kahneman and others argue challenges the idealized rationality of **homo economicus**.

So, while homo economicus assumes perfect rationality, the acknowledgment of biases shows a more adaptive, yet bounded, form of rationality that aligns better with how people actually behave in business and everyday life.

Yes, if people had "super brains" akin to the **homo economicus** model—able to process all information perfectly, without time or cognitive constraints—**they would likely not need biases**. Biases exist because they help us make decisions efficiently in the face of uncertainty, limited time, and incomplete information. A "super brain" with unlimited cognitive capacity and perfect information would have no need for shortcuts like biases. Instead, it could always compute the optimal decision without relying on mental heuristics."